

ESP32-Based Dual-Connectivity Data Logger for Continuous Environmental Monitoring for AI Applications

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Abstract

Pollution, global warming, and natural disasters are some of the current environmental problems facing humanity. Appropriate technologies are introduced to effectively solve environmental problems, and conditions are regularly monitored to collect reliable data. The Internet of Things (IoT) has revolutionized environmental protection systems by enabling real-time monitoring, automation, and data-driven decision-making. Present data loggers rely on one kind of communication, either Wi-Fi or Global System for Mobile Communications (GSM) technology, for IoT solutions. Wi-Fi offers high data rates with minimal latency; however, its performance is limited to a confined coverage area, making it unsuitable for long-distance or mobile applications. In contrast, GSM provides wide-area coverage and reliable connectivity in remote regions but suffers from slower data speeds, higher latency, and greater energy consumption. In Artificial Intelligence (AI) applications, continuous data acquisition is vital to ensure accurate training, reliable predictions, and effective automated decision-making. This paper introduces the ESP32 microcontroller-based dual-connectivity data logger, combining GSM and Wi-Fi communication in order to enable smooth data acquisition and transmission of environmental parameters for AI-based IoT applications with reduced data losses. This hybrid method will improve the performance of AI-based models in predictive maintenance, environmental monitoring, and smart agriculture by ensuring that data flows continuously and is of high quality. The experimental results showed that, when using Wi-Fi, the ESP32 delivers rapid, uniform data transfer with an elevated success rate, while GSM connectivity preserves logging functionality during Wi-Fi outages or fluctuations. Overall, the proposed dual-connectivity logger achieved a 92% success rate, reduced latency by 400 ms, and maintained moderate power consumption, proving its superiority in data reliability, accessibility, and operational efficiency. The findings show that dual connectivity has an advantage in terms of data reliability, access, and efficacy.

1 Introduction

Climate change has emerged as a significant issue facing the world today. The entire global ecosystem has significantly worsened due to climate change. Air pollution has surpassed hazardous levels in numerous countries around the world. The issue of drinking water might escalate in the future; forests are burning, glaciers are melting, and all this is occurring because of climate change. If all these things are not controlled in time, then the coming time can become even worse. The quality of air is largely influenced by a number of factors, including environmental ones, and in this case, the most relevant are particulate matter (PM_{2.5} and PM₁₀), nitrogen oxides (NO_x), carbon monoxide (SO₂), and sulfur dioxide (CO) [1]. Most of these pollutants stem from industries, vehicle emissions, and wildfires. Indirectly, these activities disrupt the ecosystem and can develop several respiratory and cardiovascular diseases [2].

Temperature plays a major role in pollution because higher temperatures both increase the amount of ground-level ozone (O₃) present and intensify the effects of pollution. Moreover, air circulation tends to fan the pollution over wider areas during heat waves, which actually causes pollution to increase [3].

Humidity also affects air pollution; high humidity helps settle dust but can increase mold spores and volatile organic compounds (VOCs), while low humidity keeps pollutants suspended in the air, increasing inhalation risks. Atmospheric pressure has an effect on pollution dispersion; smog and other pollutants are trapped around areas of marked high pressure whilst areas of marked low pressure and storms help to disperse them [4].

Wind speed can also improve air quality as a result of dispersing pollutants and aid in the circulation of polluted air and dust from other areas, thus further deteriorating the health of individuals with respiratory diseases [5]. Rainfall generally reduces pollution by washing away airborne particles, but excessive rain can contribute to water pollution by carrying chemicals, heavy metals, and waste into water bodies [6]. Figure 1 shows the different types of pollution.

Pollution and the environment are closely related. To improve sustainability and put into practice efficient pollution control measures, it is essential to comprehend their relationship. These variables allow us to forecast pollution levels and take appropriate action to lessen their negative effects on the environment and human health. We need an environmental monitoring system that can work well in urban areas as well as rural areas. Environmental monitoring is essential for tracking and managing the impact of human activities on natural ecosystems, public health, and climate stability.

Environmental monitoring is critical for risk management and disaster forecasting in the context of climate change. Scientists are able to forecast extreme weather conditions such as hurricanes, floods, and droughts by studying temperature fluctuations, rainfall, and greenhouse gas emissions. This makes possible early warning systems and disaster preparedness. Figure 2 indicates the development of environmental technology globally. The market for environmental testing is expanding due to increasing public concern about environmental degradation and the subsequent strict enforcement of environmental protection legislation [7].

The Internet of Things (IoT) is a new technology that allows numerous virtual and physical devices to link and interact with one another via the internet without requiring human involvement. IoT based dataloggers play the crucial role to collect the information from the remote area. IoT devices detect and collect various types of information about our physical environment which is subsequently utilized to develop new technologies that address issues and enhance our lives [8]. The Internet of things has achieved significant success worldwide, with billions of dollars of devices being sold and utilized across various consumer markets [9]. Figure 3 illustrates the increase of IoT devices over the past decade.

Data-based decision-making has been transformed by two new technologies, namely artificial intelligence (AI) and the Internet of Things (IoT), in number of areas, such as healthcare, smart cities, industrial automation, agriculture, and environmental monitoring [11]. These applications depend on reliable and efficient data-logging systems to support AI models that allow for real-time analysis, predictive insights, and automated responses. However, extreme difficulty to maintain constant data availability and transmission in different environments conditions [12].

Currently, most data acquisition systems depend on a single technology, such as Wi-Fi or GSM, but this is frequently insufficient for contemporary IoT applications. Wi-Fi is quick and features minimal lag, but it operates optimally only within a confined range. While GSM has better coverage area, slower speed and more energy consumption.

In AI-based applications, the demand for continuous data acquisition is even more critical, as uninterrupted input is essential for accurate model training, reliable trend prediction, and effective automated decision-making. Any interruption or loss of data can significantly reduce the reliability of the outcomes. To address this issue, the study proposed a dual connectivity datalogger which offers a robust solution by enabling devices to switch seamlessly between Wi-Fi and GSM, thereby ensuring data integrity and consistent analytical performance.

This paper describes the design and operation of a dual-connectivity environmental monitoring and logging system. The proposed system consists of the ESP32 microcontroller, GSM module, BME680 sensor, anemometer, and ThingSpeak cloud platform. The ESP32 microcontroller supports dual-core, low power consumption, and affordability [13]. We can utilize one core of the ESP32 microcontroller for Wi-Fi connectivity and the other for GSM connectivity for round-the-clock data transmission. For low power consumption and high speed, we can use Wi-Fi connectivity, and we have the option to utilize GSM connectivity when Wi-Fi is unavailable or in remote areas. For this operation, FreeRTOS has been used that provides multitasking support for the ESP32 dual-core microcontroller. The dual connectivity of the ESP32 microcontroller not only diminishes the data loss but also improves the quality and accessibility of the data. This allows us to achieve the smooth and continuous data flow necessary for AI applications and real-time decision-making. The system provides a flexible and secure solution for a variety of AI and IoT-based environmental monitoring applications via GSM and Wi-Fi communication. This technology can make it possible to monitor changes in the weather from anywhere at any distance. The system can be a highly advanced, efficient, accurate, cost-saving, and reliable weather station that will benefit any person who wishes to track environmental changes on a regular basis.

2. Literature Review

Monitoring of the environment is critical for balancing human interactions with nature, advancing novel technologies, and safeguarding public health. Today, the logging devices, with their capability of uninterrupted and precise substitution recording the environmental parameters pertaining to air like its quality, temperature, humidity, atmospheric pressure, and wind speed, came to the rescue. Based on their function, these logging devices could be divided into wired and wireless categories [14]. Fig. 4 shows the different types of dataloggers and their applications. Directly connected wired loggers guarantee high-quality data collection and for this reason are best suited for industrial-scale pollution monitoring processes, laboratory-controlled experiments, and studies of environmental systems. Various types of USB interfaces, Ethernet, and even serial communication protocols are familiar methodologies for sending data collected to a central system. The main drawback, however, remains the cables on which these loggers are dependent, which limits their deployment options for remote locations [21].

Contrary to wired loggers, wireless loggers transmit data directly and in real time using diverse technologies, such as Wi-Fi, Bluetooth, LoRa, Zigbee, or even GSM. These loggers are commonly applied for smart cities, weather stations, agriculture, and other forest monitoring. These solutions can overcome significant hurdles and allow constant overseeing of changes taking place in the environment in inaccessible places.

Gupta et al. investigated an IoT-based data collection system and smart switch to manage the data acquisition system. The system was efficient, reliable, and suitable for usage in the outdoors. It conserved 58% of energy and prolonged the life of sensors [14]. Kodali et al. used sensors linked to cloud services and WeMo to track the weather [15]. Saini et al. connected a PC to a graphical user interface (GUI) using ZigBee; they downloaded data from the internet and showed it on an LCD screen [16]. Gupta et al. analyzed the effect of dust on the performance of the PV system with the use of a LABVIEW-based data logger [17]. Palle et al. used the CC3200, a simple Link Wi-Fi internet chip, to detect temperature and humidity [18]. Gupta et al. developed a cost-effective PV analyzer using a sequential design comprising voltage, current, temperature, rheostat, and Arduino UNO sensors for measuring and recording output parameters [19]. Shaout et al. measured three meteorological variables: wind direction, wind speed, and temperature. The LCD received the information [20]. Using a variety of sensors, such as an ESP32 Wi-Fi module, DHT11 temperature and humidity sensor, anemometer, dust sensor, and ThingSpeak cloud platform, Gupta et al. created an open-source wireless data acquisition system [21]. An Arduino Uno-based system was utilized by Amber Katyal et al. to determine whether the environment had light, cloudy weather [22]. Gupta et al. proposed the IoT-based data acquisition system using ESP8266 for monitoring the battery performance in electric vehicles [23]. Raj et al. developed a LabVIEW-based PV analyzer. The system consists of a DAQ card, a temperature sensor, a voltage sensor, and a current sensor [24].

TABLE-1 List of existing IOT based Data logger

Author	Microcontroller	Connectivity	Transmission by	Purpose
Touati et al 2018 [28]	Arduino Mega 2560	Wireless	XBee Pro 1	For PV system
Mhetre et al. 2019 [29]	Arduino Uno	Wireless	Zigbee network	For PV system
Rezk et al. 2017 [30]	NI USB-6009	Wired	Ethernet	For PV system
Paredes-Parra et al. 2019 [31]	ATmega328P	Wireless	LoRa	For PV system
Payne et al. 2019 [32]	Atmega-328	Wireless	GSM	For PV system
Olivier et al. 2020 [33]	Arduino Uno., ESP8266	Wireless	Wi-Fi	For PV system
Aghenta al. 2019 [34]	Arduino Uno	Wired	Ethernet	For PV system
Bikrat et al. 2019 [35]	Raspberry Pi 3	Wireless	Bluetooth	For PV system
Badave et al. 2018 [36]	CC3200 launchpad (ARM)	Wireless	Wi-Fi	For PV system
Singh et al. (2022) [37]	ESP8266	Wireless	Wi-Fi	For weather monitoring system
Nagane et al. (2022) [38]	ESP8266	Wireless	Wi-Fi	For weather monitoring system
Roopa M L et al. (2021) [39]	ESP8266	Wireless	Wi-Fi	For weather monitoring system
Abbas, M. (2022) [40]	a Raspberry Pi 3	Wireless	Wi-Fi	For weather monitoring system
Karthigaen (2020) [41]	Arduino Uno, ESP8266	Wireless	Wi-Fi	For weather monitoring system

In a model presented by R. Kavin et al., sensors are configured to monitor variables such as temperature, humidity, and CO value, and information is collected via the IDE (Integrated Development Environment). The outcome evaluation will be sent via Wi-Fi to the final client. An ATmega 328 controller manages the devices during each detection phase, gathers sensor data, and transmits it to end users through the cloud [25]. A wireless and Internet of Things-based weather observation system was proposed by Mohit Tiwari et al. [26]. Tellawar et al. developed a data logger to monitor parameters using an ESP32 microcontroller, store them on a micro-SD card, and show data in real time through the Blynk app [27]. Table 1 demonstrates some existing data loggers.

After conducting a detailed literature survey, it was observed that most existing data loggers rely solely on Wi-Fi as their primary communication technology, but this approach is often insufficient for contemporary IoT applications with AI. Wi-Fi offers high data rates with minimal latency; however, its performance is limited to a confined coverage area, making it unsuitable for long-distance or mobile applications. In contrast, GSM provides wide-area coverage and reliable connectivity in remote regions but suffers from slower data speeds, higher latency, and greater energy consumption.

For AI-based applications, the requirement becomes even more stringent, as continuous data receiving is crucial to maintain accurate model training, trend prediction, and automated decision-making. Any gaps in data can compromise the reliability of insights. Hence, dual connectivity provides a practical solution by allowing devices to switch intelligently between Wi-Fi and GSM without affecting data integrity or analytical performance.

To address this issue, the present study proposes a dual-connectivity data logger that utilizes a single ESP32 microcontroller with an integrated dual-core architecture. This allows one core to handle Wi-Fi communication while the other manages GSM communication, thereby achieving reliable dual connectivity at reduced cost and hardware requirements. These hybrid or dual-connectivity solution with one microcontroller can combine the strengths of both technologies, ensuring reliable, energy-efficient, and continuous data transmission in diverse environments with low cost.

3. System Description and Methodology

The proposed dual-connectivity data logger is designed for harsh environments and long-term data collection of environmental parameters such as temperature, humidity, pressure, air quality index, and wind speed in the various regions where Wi-Fi is available and for remote regions where Wi-Fi infrastructure is limited. This system consists of the ESP32 microcontroller, GSM module, BME680 sensor, anemometer, and ThingSpeak cloud platform. Fig. 5 shows the block diagram of the proposed dual-connectivity data logger. Table 2 shows the list of components used in the proposed dual-connectivity data logger. Table 3 shows the specifications of components used in the proposed dual-connectivity data logger.

The ESP32 microcontroller supports both GSM and Wi-Fi, but due to its larger data rate and affordability, Wi-Fi is favoured. We set the ESP32 microcontroller in such a way that Core 1 performs GSM connectivity for data exchange and Core 0 manages Wi-Fi communication. FreeRTOS-based multitasking is used to effectively manage multiple tasks that are running concurrently on both cores. The first core was made for Wi-Fi connectivity, and the second one was made for GSM connectivity. This ensures that WiFi and GSM operations are operated simultaneously without conflict, enhancing efficiency and reactivity in IoT applications with AI. The subsections that follow present the implementation of the hardware and software components.

• Hardware Implementation

The ESP32 is ideal for hardware setups because of the low cost and built-in Wi-Fi. A single GSM module ensures reliable connections. The main hardware components include:

- **ESP32 Development Board:** This board enables communication through dual-core processing.
- **GSM Module (e.g., SIM800):** This module provides backup communication over cellular networks and works well in nearby areas.
- **BME680 sensor and anemometer Sensor:** These components ensure the data collection of environmental parameters such as temperature, humidity, pressure, air quality index, and wind speed.
- **Power Supply:** A reliable and stable power supply is essential for the proper functioning of the ESP32 microcontroller, the BME680 environmental sensor, and the wind sensors.

- **Software Implementation**

The Software will be capable of supporting ESP-IDF framework and FreeRTOS. Modular codes enumerated below will be additional to the following functions:

- **Wi-Fi Connectivity Handler:** Offers connectivity and sets up the Wi-Fi network; pings the API server every few seconds.
- **GSM Connectivity Handler:** This handler is used to offer a backup option for data transmission when Wi-Fi is not available.
- **Cloud Integration Module:** The following module uses RESTful APIs of ThingSpeak, to convey data to cloud storage.

To facilitate continuous switching, the Task Scheduler (FreeRTOS) executes parallel tasks, including keeping track of the network status.

- **Cloud Integration and Data Collection**

After the ESP32 collects data from the connected sensors, the information is transmitted to a cloud backend for storage and further processing. Free-tier cloud platforms, such as ThingSpeak, are utilized for this purpose. Data transfer is carried out through RESTful API calls, ensuring secure, reliable, and efficient communication with the cloud. The ThingSpeak cloud platform stores and displays the parameters. The platform's integrated graphical visualization tools allow users to monitor sensor data in real time. It ensures consistent and reliable monitoring of environmental parameters. All information can be accessed from anywhere at any time. ThingSpeak Cloud provides the MATLAB tools that enable AI and machine learning applications such as anomaly detection, trend analysis, and predictive modelling. This makes it possible for our data collection system to function more efficiently.

The task segmentation for cores optimizes the assignment of resources for ongoing real-time data capture and sending. The proposed data logging system defaults to GSM connectivity for data transmission in the absence of Wi-Fi. The ESP32 is continuously reading environmental data from sensors. Based on network availability, the data is processed and sent to the ThingSpeak cloud platform over GSM or Wi-Fi. This approach proposes continuous data collection from remotely located APIs and simultaneous storage in a cloud-based backend, even in the face of a network failure. This deployment demonstrated a reliable and efficient IoT solution that enabled continuous data logging for environmental monitoring.

To coordinate multiple tasks being executed in parallel on both cores effectively, FreeRTOS-based multitasking is employed. Core 0 handles the Wi-Fi task, i.e., data uploads as well as Over-The-Air (OTA) updates. Core 1 checks the status of the GSM network and ensures cellular data transmission whenever needed. A failover system saves the data in an SD card until the link is reestablished in case both network links are lost. Fig.6 shows experimental setup and the proposed dual-connectivity data logger. This method guarantees a reliable, effective, and fault-tolerant IoT data logging system that runs continuously in various environments by enabling real-time monitoring, predictive analytics, and a smooth connection with cloud services. Figure 7 presents the flowchart of the suggested dual-connectivity data logger.

TABLE.2 Components of Proposed Dual-Connectivity Data Logger

Components	Hardware/Software	Remark
ESP32 Micro controller	Hardware	Wi-fi Module
GSM Module	Hardware	To provide GSM Connectivity
BME680 Sensor	Hardware	To sense Temperature, Humidity, Pressure and Air quality Index
Anemometer	Hardware	To Sense Wind Speed
SD card	Hardware	To storage data
Arduino IDE	Software	For coding
ThingSpeak	-	Cloud Platform

TABLE.3 Components and its Specifications

Component	Parameters	Specification
ESP32 Wi-Fi Module	Operating voltage	3.3 V or 5V
	Available GPIO Pins	Up to 34 GPIOs
	RAM	520KB SRAM, up to 16MB PSRAM
	Clock speed	240 MHz
	Processor	Dual-core Xtensa LX6 @ 240MHz
GSM Module (Sim900A)	Operating Voltage	3.5 V- 4.5 V
	Operating Current	Idle Mode:10mA, Active mode: 100- 200mA, Peak transmission: 1.6 A
	Operating Temperature	-40°C to 85°C
	Control Interface	AT Commands via UART
	Frequency Bands	GSM 900/1800 MHz (Dual-band)
BME680 Sensor	Measurement range	0-100%RH-40°C-85°C, 300-1100hPa
	Accuracy	±0.5°C, ± 3%, ± 0.25 %
	Operating Voltage	3.3V
	Average Current	2.1 µA
SD Card	Type	Micro SD Card
	Capacity	4GB – 32GB (FAT32)
	Interface	SPI Mode
	Temperature Range	-25°C to 85°C
	Operating Voltage	3.3V (direct connection)
Anemometer	Supply Voltage:	7 to 24 V DC
	Max wind Speed	70m/s
	Accuracy	1m/s
	Supply Voltage:	7 to 24 V DC

4. Results and Discussion

To evaluate the reliability of the proposed dual-connectivity data logging system, we used a BME680 Sensor, wind sensor, to measure environment's humidity and temperature, pressure, gas resistance, air quality index and wind speed. The system was intended to function with dual-network connectivity, using

GSM as a fallback in case Wi-Fi failed and Wi-Fi as the main communication channel. Three factors were taken into consideration when assessing the system's performance: success rate, latency and power consumption.

1. **Success Rate (%)** – Measures how reliably data is transmitted.

Success Rate (%) = [Number of Successful Received data on ThingSpeak Cloud x 100 / Total Number of Received data on Serial Monitor]

2. **Latency (ms)** – Measures transmission delay.

Latency (ms) = [Time of Acknowledgement data on ThingSpeak Cloud – Time of Received data on Serial Monitor]

3. **Average Current (mA)** – Indicates power consumption.

Average Current (mA) = [Total Current Consumed (mAh) x 1000 / Total Time (h)]

Fig.8 shows that the ESP32 microcontroller with Wi-Fi connectivity transmits data quickly and consistently, resulting in a higher success rate, whereas GSM connectivity ensures that data is logged even in areas with intermittent or no Wi-Fi. To ensure uninterrupted data logging, an automatic failover system transitioned the connection from Wi-Fi to GSM whenever a Wi-Fi disconnection was identified. Fig.9 shows serial monitor's screen while sending data by Wi-Fi connectivity. Fig.10 shows the switching from Wi-Fi to GSM connectivity. Fig.11 shows serial monitor's screen while sending data by GSM connectivity.

The average latency observed across different modes highlights a clear trade-off between speed and reliability. As illustrated in Fig. 12, the average latency in Wi-Fi mode is approximately 300 ms, in GSM mode around 600 ms, and in hybrid mode nearly 450 ms. This demonstrates that while Wi-Fi provides the lowest latency, its dependency on signal availability may limit its applicability in remote or infrastructure-scarce regions. Conversely, GSM connectivity, despite exhibiting higher latency, ensures continuous data logging in areas where Wi-Fi networks are absent or unreliable. Hybrid mode, with an average latency of 450 ms, strikes a balance between the two, ensuring both timeliness and resilience in data delivery.

Further analysis in Fig. 13 reveals the success rate of data transmission across connectivity modes. Wi-Fi mode achieves the highest consistency with a 96% success rate, reflecting its stability in environments with strong network coverage. GSM mode, although limited in speed, provides a commendable 88% success rate, ensuring that critical environmental data is not lost in regions with limited Wi-Fi access. Importantly, the hybrid mode achieves a 92% success rate, balancing both connectivity options to provide robust and seamless data transmission. This hybrid strategy is especially significant in ensuring uninterrupted data flow for environmental monitoring systems where gaps or delays in data collection can compromise downstream analysis.

The proposed data logger is based on IoT that can send data from a remote location, so power consumption is crucial. When connected to Wi-Fi, the device used 220 mA and had a 45-hour operating time. It drew 1.6 A and ran for 6.25 hours in GSM mode. However, when dual connectivity was enabled, the system's battery life varied from 20 to 5.5 hours, and it drew approximately 1.8 A, depending on how long Wi-Fi was unavailable. The average current and battery runtime for each mode is shown in fig 14 and fig 15 respectively. Although it consumes more energy than Wi-Fi mode, the proposed dual-connectivity data logger receives data without any interruption. Table.5 shows the comparison of ESP32 based data logger with Wi-Fi and GSM connectivity. Fig.16 compares the performance of three communication modes: Wi-Fi, GSM, and Hybrid Mode (Wi-Fi + GSM) across four parameters.

TABLE.5 Comparison of ESP32 based datalogger with Wi-Fi and GSM connectivity

Parameters	ESP32 (Wi-Fi)	ESP32 + GSM (SIM800/900)
Connectivity	Reliable in urban areas with Wi-Fi availability.	Effective in urban & remote areas with cellular coverage.
Data Transfer Speed	High (up to 150 Mbps).	Slower (up to 85.6 kbps with GPRS).
Power Consumption	Low	High
Cost	Lower due to built-in Wi-Fi.	Higher due to GSM module and SIM plan.
Suitability	Ideal for urban and semi-urban areas.	Best for rural and off-grid locations.

This hybrid approach becomes particularly vital in the context of AI-driven environmental monitoring applications, where seamless data transmission is essential. For AI models to generate accurate predictions, continuous and reliable streams of data are indispensable. Hybrid connectivity ensures that sudden interruptions in one network do not compromise the integrity of the dataset, thereby supporting uninterrupted analytics and real-time decision-making. In essence, the hybrid mode not only improves robustness in diverse field conditions but also lays a strong foundation for scalable, intelligent environmental monitoring systems.

5. Future Work

Although the present work successfully demonstrates the implementation of dual connectivity (Wi-Fi and GSM) for reliable data logging, there are several promising directions for future research and development:

- i. *Advanced AI Integration:* Future research might use deep learning and real-time machine learning algorithms on the ESP32 or edge devices to log, analyze, and forecast environmental parameters locally. This would improve decision-making latency and lessen reliance on cloud servers.

- ii. *Seamless Hybrid Switching Mechanism:* Further optimization of adaptive switching algorithms based on signal strength, energy consumption, and network availability could improve system robustness and ensure near-zero data loss for critical applications, even though the current hybrid mode increases success rate and reliability.
- iii. *Energy Optimization:* Adding low-power protocols (like LoRa, NB-IoT, or MQTT-SN) and investigating renewable energy sources (like solar-powered ESP32 nodes) could prolong the deployment time in isolated or off-grid areas.
- iv. *Data Security and Privacy:* Ensuring data confidentiality and integrity, particularly in AI-based applications, will require integrating secure authentication methods and lightweight encryption designed for IoT microcontrollers.
- v. *Application-Specific Adaptations:* In other fields where continuous data transfer is essential, like smart agriculture, electric vehicle monitoring, industrial IoT, and disaster management, the suggested dual connectivity model can be expanded.

6. Conclusion

The environment is currently threatened by issues like pollution, global warming, and natural disasters. The monitoring and data logging of environmental parameters are required for risk management and disaster prediction. The stored data is very useful for scientists to predict extreme weather events like hurricanes, floods, and droughts. This allows for early warning systems and disaster preparedness. Existing data loggers depend on a single type of communication, either Wi-Fi or GSM communication technology, for IoT applications. Wi-Fi is quick and features minimal lag, but it operates optimally only within a confined range. while GSM has a better coverage area, slower speed, and more energy consumption. The proposed dual-connectivity data logger on the ESP32 microcontroller for continuous data transfer was successfully implemented by combining Wi-Fi and GSM communication. The proposed dual-connectivity data logging system captured environmental parameters such as temperature, humidity, pressure, air quality index, and wind speed without interference, even in poor network conditions. The results are shown by the ESP32 microcontroller with Wi-Fi connectivity to transmit data quickly and consistently, resulting in a higher success rate, whereas it is ensured by GSM connectivity that data is logged even in areas with intermittent or no Wi-Fi. The proposed dual-connectivity data logger achieved a 92% transmission success rate, reduced latency by 400 ms, and maintained moderate power consumption. The benefits of dual connectivity in terms of data reliability, accessibility, and effectiveness are demonstrated by the results. The performance of AI models in predictive maintenance, environmental monitoring, and smart agriculture is improved by this hybrid approach through the ensuring of continuous and high-quality data flow.

Declarations

Disclosure of interest:

No potential competing interest was reported by the authors.

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Author Contribution

Eshita Gupta & Vinay Gupta contributed to the conceptualization, design, and theoretical framework of the study. Eshita Gupta, Vinay Gupta & Sanskar Bansal were responsible for data collection, software implementation, methodology development and initial manuscript drafting. Vikas Chaudhary and Rupendra Kumar Pachauri contributed to, data analysis, and manuscript review.

Data Availability

The datasets used in the current study are available from the corresponding author upon reasonable request.

References

1. Oliveira, F., Costa, D.G., Assis, F., Silva, I.. Internet of Intelligent Things: A convergence of embedded systems, edge computing and machine learning. *Internet of Things*, 26, 101153. (2024).
2. Jayanthi, S., Srividhya, E., Ramshanka, N., Srinivasan, T.R., Prabhakar, P.B.E.: Cloud-based weather station using IoT technology to track the air parameters. In: *2023 Second International Conference on Electronics and Renewable Systems (ICEARS)*, pp. 537–541 (2023).
3. Qasim, H.H., Jasim, A.M., Hashim, K.A.: Real-time monitoring system based on integration of Internet of Things and Global System of Mobile using Raspberry Pi. *Bull. Electr. Eng. Inform.* 12(3), 1418–1426 (2023).
4. Merz, B., et al.: Impact forecasting to support emergency management of natural hazards. *Rev. Geophys.* 58(4), (2020).
5. Kumar, P., et al.: An overview of monitoring methods for assessing the performance of nature-based solutions against natural hazards. *Earth-Sci. Rev.* 217, 103603 (2021).
6. Tey, F.J., Wu, T.-Y., Chen, J.-L.: Machine learning-based short-term rainfall prediction from sky data. *ACM Trans. Knowl. Discov. Data* 16(6), 1–18 (2022).
7. <https://www.thebusinessresearchcompany.com/report/environmental-technology-global-market-report>.

8. Aghenta, L.O., Iqbal, T.: Design and implementation of a low-cost, open-source IoT-based SCADA system using ESP32 with OLED, ThingsBoard, and MQTT protocol. *AIMS Electron. Electr. Eng.* 4(1), 57–86 (2019)
9. Alahi, M.E.E., Sukkuea, A., Tina, F.W., Nag, A., Kurdthongmee, W., Suwannarat, K., Mukhopadhyay, S.C.: Integration of IoT-Enabled Technologies and Artificial Intelligence (AI) for Smart City Scenario: Recent Advancements and Future Trends. *Sensors* 23(11), 5206 (2023)
10. Atlam, H.F., Alenezi, A., Alassafi, M.O., Wills, G.B.: Blockchain with Internet of Things: Benefits, Challenges, and Future Directions. *Int. J. Intell. Syst. Appl.* 10, 40–48 (2018)
11. Thompson, J., et al.: AI-driven IoT devices: Applications and opportunities. *AI Robot. Today* 3, 45–57 (2023)
12. Chen, Y., et al.: Cloud-assisted AI processing for embedded systems. *IEEE Internet Things J.* 9, 1254–1267 (2022)
13. Espressif: Programming ESP32 for cloud-based applications. Espressif Blog
14. Gupta, V., Sharma, M., Pachauri, R.K., Babu, K.N.D.: A low-cost real-time IoT-enabled data acquisition system for monitoring of PV system. *Energy Sources Part A Recover Util. Environ. Eff.* 43(20), 2529–2543 (2021)
15. Kodali, R.K., Mandal, S.: IoT based weather station. In: 2016 IEEE International Conference on Control, Instrumentation and Computational Technologies (ICCICCT), pp. 1–10 (2016)
16. Saini, H., Thakur, A., Ahuja, S., Sabarwal, N., Kumar, N.: Arduino based automatic wireless weather station with remote graphical application and alerts. In: 2016 IEEE International Conference on Signal Processing and Integrated Networks (ICSPIN), pp. 605–609 (2017)
17. Gupta, V., Raj, P., Yadav, A.: Investigate the effect of dust deposition on the performance of solar PV module using LABVIEW based data logger. In: 2017 IEEE Int. Conf. Power Control Signals Instrum. Eng. (ICPCSI), Chennai, India, pp. 742–747 (2017)
18. Palle, D., Kommu, A., Kanchi, R.R.: Design and development of CC3200-based cloud IoT for measuring humidity and temperature. In: 2016 IEEE International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT), pp. 3116–3120 (2016)
19. Gupta, V., Raj, P., Yadav, A.: Design and cost minimization of PV analyzer based on Arduino UNO. In: 2017 IEEE Int. Conf. Power Control Signals Instrum. Eng. (ICPCSI), Chennai, India, pp. 1337–1342 (2017)
20. Shaout, A., Li, Y., Zhou, M., Awad, S.: Low-cost Embedded weather station with intelligent system. In: 2014 10th IEEE International Computer Engineering Conference (ICEC), pp. 100–106 (2015)
21. Gupta, V., Sharma, M., Pachauri, R., Babu, K.N.D.: Performance Analysis of Solar PV System Using Customize Wireless Data Acquisition System and Novel Cleaning Technique. *Energy Sources Part A Recover Util. Environ. Eff.* 44(20), 2529–2543 (2022)
22. Katyal, A., Yadav, R., Pandey, M.: Wireless Arduino based weather station. *Int. J. Adv. Res. Comput. Commun. Eng.* 5(4), 274–276 (2016)

23. Gupta, V., Sharma, N., Maram, D., Priyadarshi, H.: IoT-enabled data acquisition system for electric vehicles. *AIP Conf. Proc.* 2294(1), 040002 (2020)
24. Raj, P., Yadav, A., Nizam, F., Gupta, V., Garg, P.: Design and Analysis of PV Analyzer on LabVIEW. In: Singh, R., Choudhury, S., Gehlot, A. (eds.) *Intelligent Communication, Control and Devices*, pp. 1135–1141. Springer, Singapore (2018)
25. Kavin, R., Lakshmi, K., Rani, S.S., Kumar, K.R.: Weather monitoring system using internet of things. In: 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS), pp. 26–29 (2021)
26. Tiwari, M., Narang, D., Goel, P., Gadhwal, A., Gupta, A., Chawla, A.: Weather monitoring system using IoT and cloud computing. *Int. J. Adv. Sci. Technol.* 29(12), 2473–2479 (2020)
27. Tellawar, M.P.: An IoT-based smart solar photovoltaic remote monitoring system. *Int. J. Eng. Res. Technol.* 8(9) (2019)
28. Touati, F., Gonzales, A.S.P., Qiblawey, Y., Benhmed, K.: A customized PV performance monitoring system in Qatar's harsh environment. In: 2018 6th Int. Renew. Sustain. Energy Conf. (IRSEC), Rabat, Morocco, pp. 1–6 (2018)
29. Mhetre, C., Narkhede, P., Pashte, Y., Patankar, N., Nishan, P.: Solar Powered Wireless Sensor Network Using Zigbee Module. In: 2019 Int. Conf. Intell. Comput. Control Syst. (ICICCS), Madurai, India, pp. 468–473 (2019)
30. Rezk, H., Tyukhov, I., Al-Dhaifallah, M., Tikhonov, A.: Performance of data acquisition system for monitoring PV system parameters. *Meas.* 10, 1008–1021 (2017)
31. Paredes-Parra, J.M., García-Sánchez, A.J., Mateo-Aroca, A., Molina-García, A.: An alternative internet-of-things solution based on LoRa for PV power plants: Data monitoring and management. *Energies* 12, 881 (2019)
32. Payne, E.K., Lu, S., Wang, Q., Wu, L.: Concept of designing thermal condition monitoring system with ZigBee/GSM communication link for distributed energy resources network in rural and remote applications. *Processes* 7, 383 (2019)
33. Olivier, M.M., Kassegne, S., Muvengei, M.: Real-time monitoring and diagnosis of organic solar cell stability. *Int. J. Sci. Technol. Res.* 9, 2385–2391 (2020)
34. Aghenta, L.O., Iqbal, M.T.: Design and implementation of a low-cost, open-source IoT-based SCADA system using ESP32 with OLED, ThingsBoard, and MQTT protocol. *AIMS Electron. Electr. Eng.* 4, 57–86 (2019)
35. Bikrat, Y., Salmi, K., Azghiou, K., Benlghazi, A., Benali, A., Moussaid, D.: Intelligent wireless system for PV supervision based on the Raspberry Pi. *Adv. Sci. Technol. Eng. Syst. J.* 4, 94–98 (2019)
36. Badave, P.M., Karthikeyan, B., Badave, S.M., Mahajan, S.B., Sanjeevikumar, P., Gill, G.S.: Health monitoring system of solar photovoltaic panel: An internet of things application. In: *Lect. Notes Electr. Eng.* 435, 347–355 (2018)
37. Singh, A.K., et al.: IOT Live Weather Monitoring System Using Nodemcu ESP8266. *Int. Res. J. Mod. Eng. Technol. Sci.* 4(5) (2022)

38. Nagane, D., et al.: IOT based Weather Reporting System. Int. J. Adv. Res. Sci. Commun. Technol., 622–624 (2022). <https://doi.org/10.48175/ijarsct5082>

39. Roopa, M.L., et al.: IOT based Real Time Weather Prediction System using NODEMCU 12-E ESP266 and Lab View. Int. J. Eng. Res. Technol. (2021)

40. Abbas, M.: IOT based weather station by using Raspberry Pi 3. IoT World. Available at: <https://iotworld.co/2018/01/iot-basedweather-station-by-using-raspberry-pi-3/>

41. Karthigaen, K., Nithyalakshmi, R. Internet of Things Based Wireless Weather Monitoring System Using Blynk Server. Int. Res. J. Mod. Eng. Technol. Sci. 2(9) (2020).

Figures

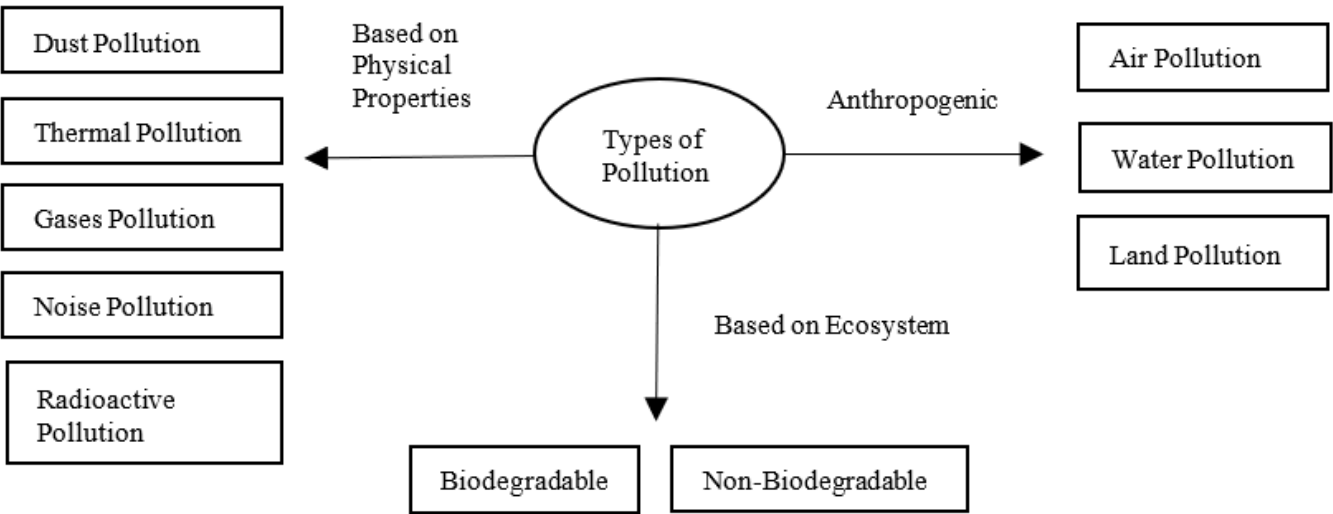


Figure 1

Types of Pollution

Environmental Technology Global Market Report 2025

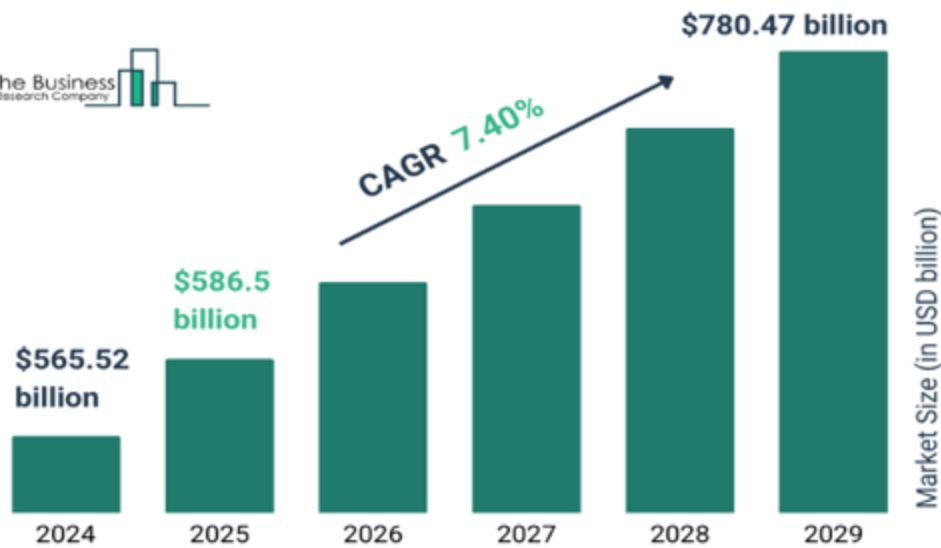


Figure 2

Growth of Environment Technology in worldwide [7]

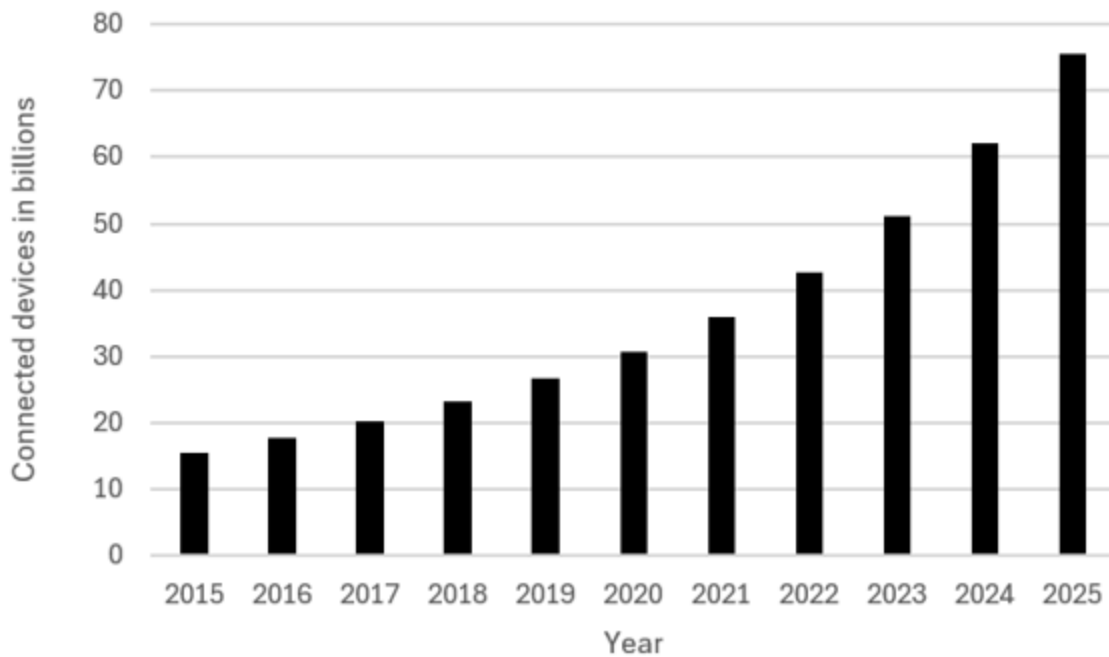


Figure 3

Growth of Internet of Things (IoT) from 2015 to 2025 [10]

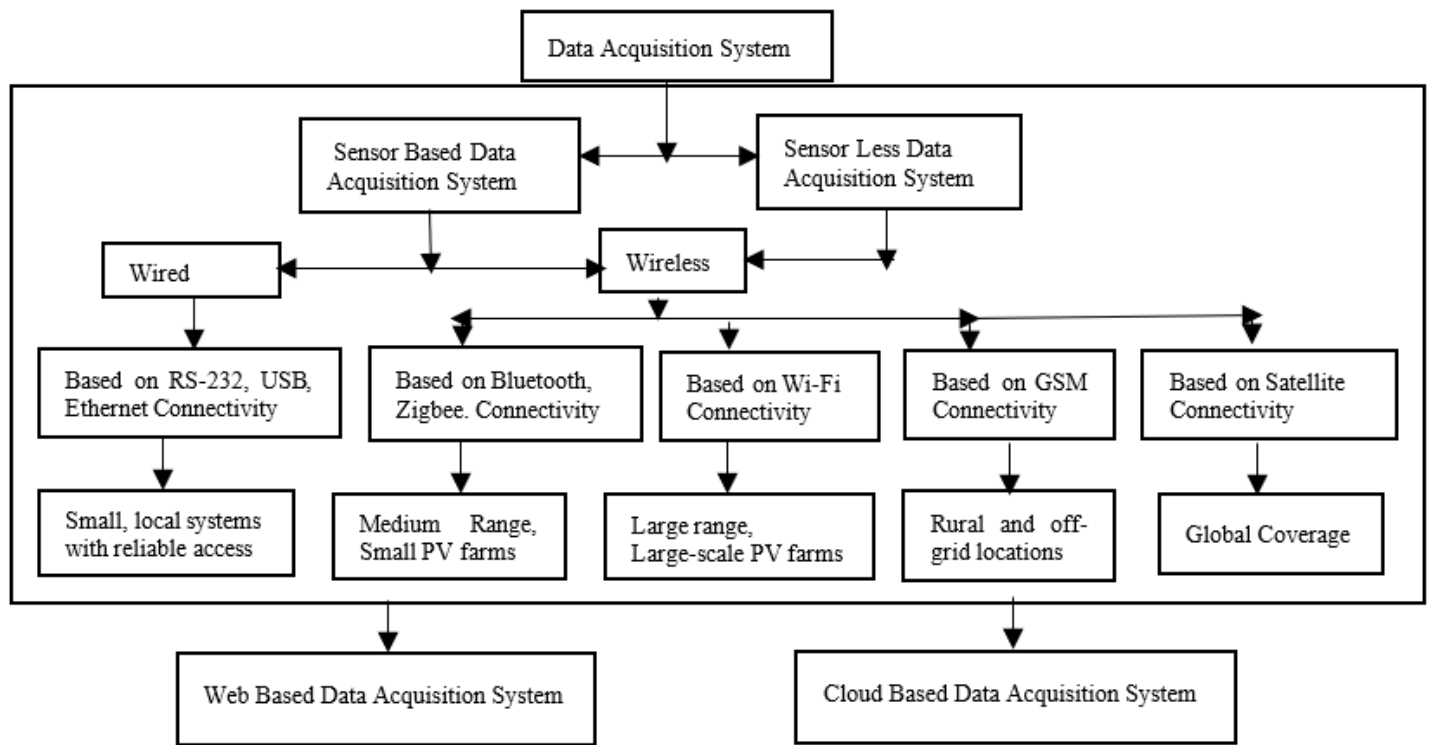


Figure 4

Types of Different Data Acquisition System

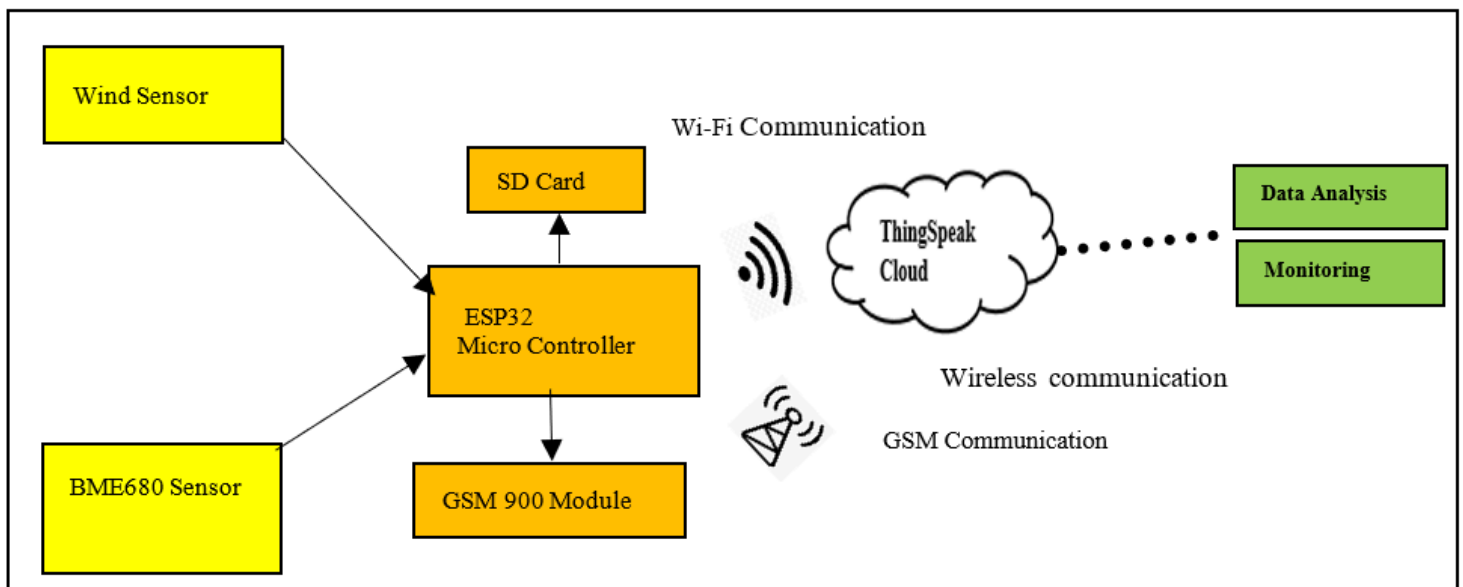


Figure 5

Block diagram of proposed dual-connectivity Data logger

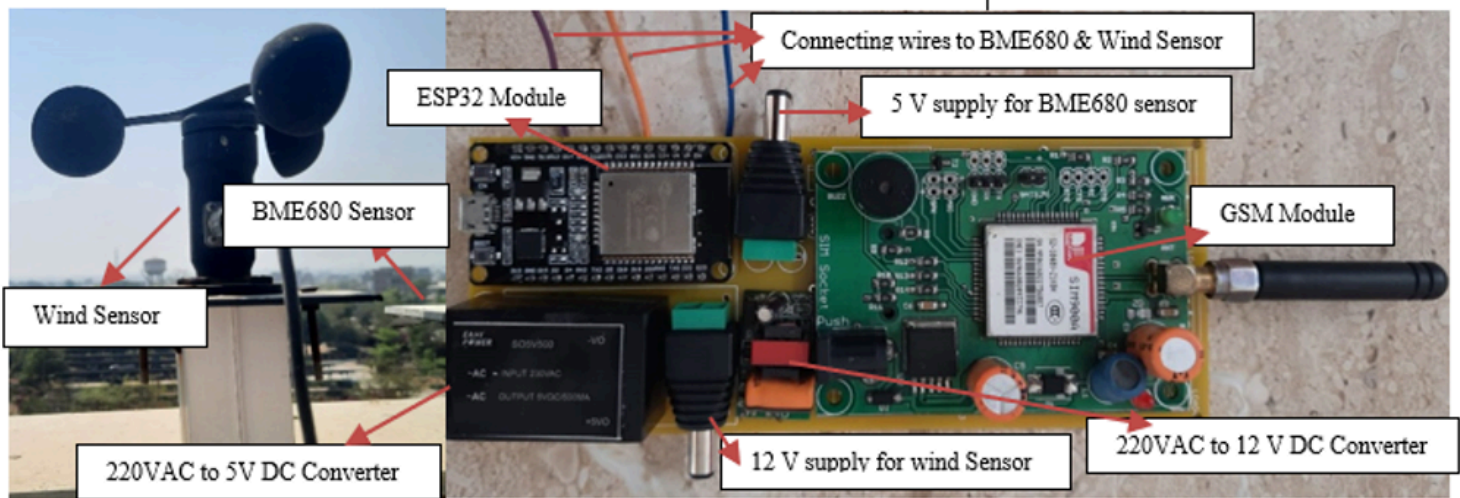


Figure 6

Experimental Setup and the proposed dual-connectivity data logger

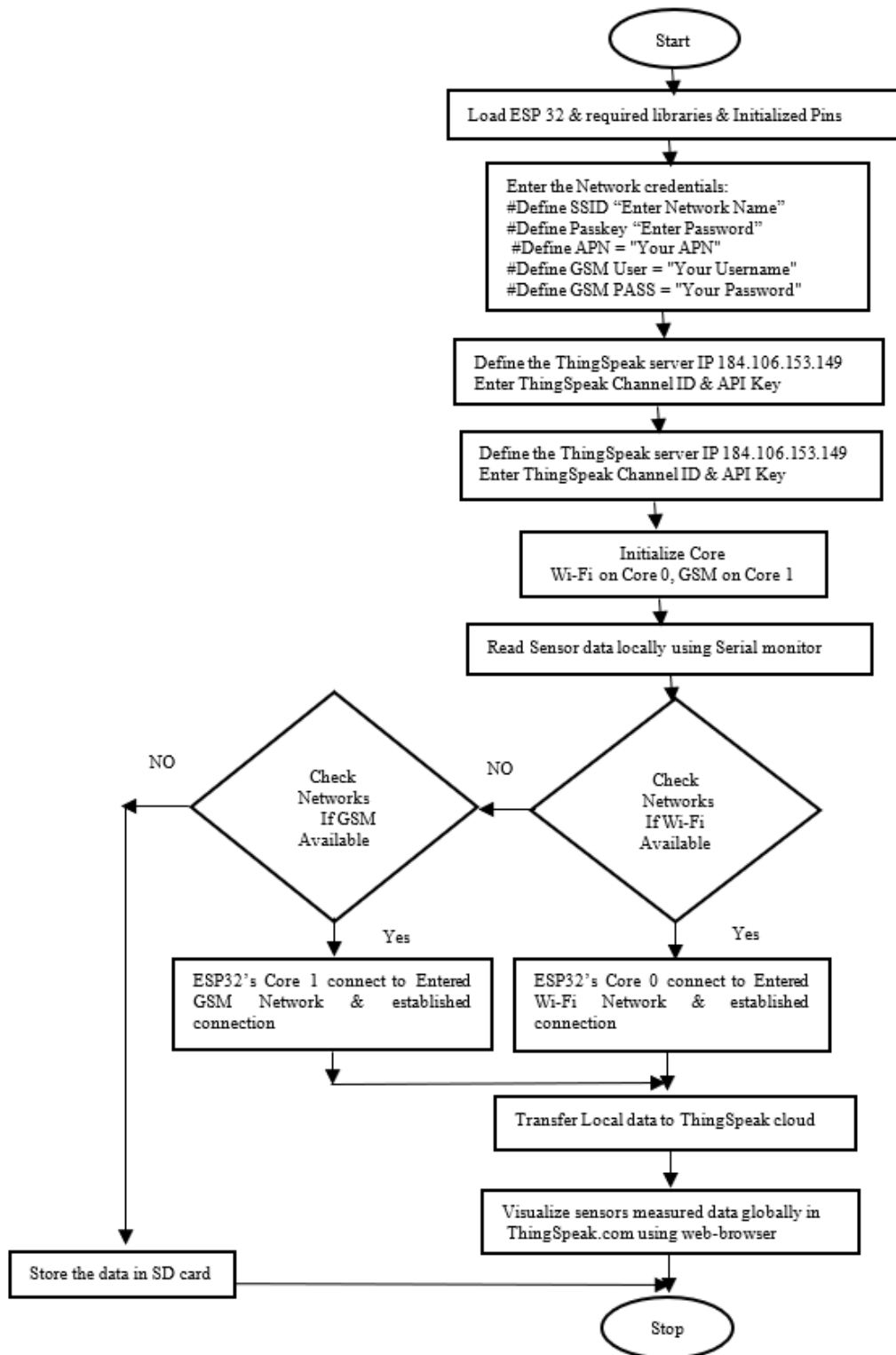


Figure 7

Flow Chart of the proposed System

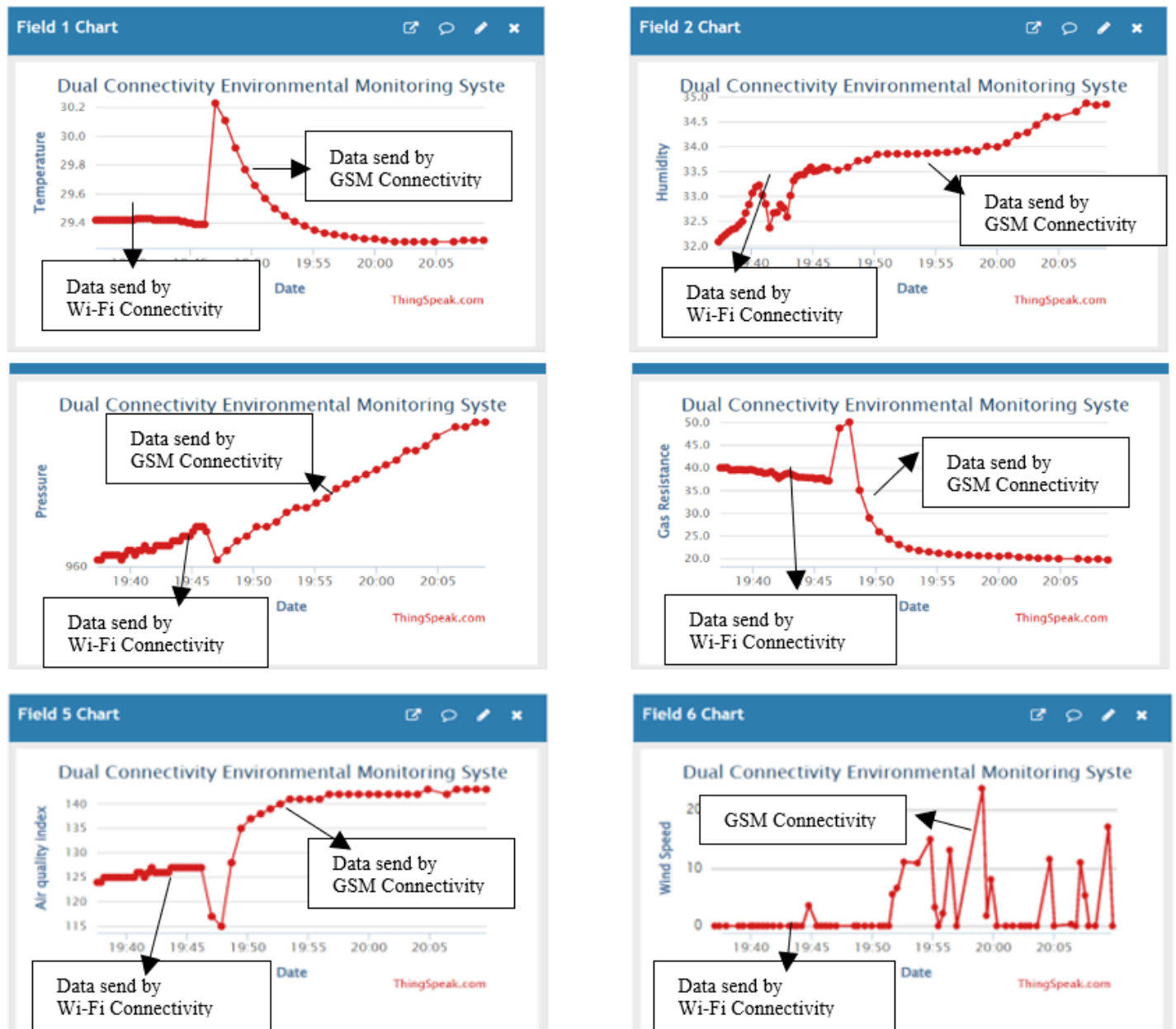


Figure 8

Data sent to ThingSpeak cloud by Wi-Fi Connectivity and GSM Connectivity

```
COM6
Temp: 29.42°C, Humidity: 32.29%, Pressure: 960.06 hPa, Gas: 39.54 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 32.34%, Pressure: 960.06 hPa, Gas: 39.54 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 32.36%, Pressure: 960.06 hPa, Gas: 39.62 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 32.43%, Pressure: 960.06 hPa, Gas: 39.62 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 32.50%, Pressure: 960.05 hPa, Gas: 39.54 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 32.67%, Pressure: 960.06 hPa, Gas: 39.54 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 32.84%, Pressure: 960.07 hPa, Gas: 39.65 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 33.07%, Pressure: 960.07 hPa, Gas: 39.47 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.42°C, Humidity: 33.19%, Pressure: 960.06 hPa, Gas: 39.17 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.43°C, Humidity: 33.23%, Pressure: 960.07 hPa, Gas: 39.14 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.43°C, Humidity: 33.03%, Pressure: 960.07 hPa, Gas: 38.81 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
Sending Data via WiFi...
Temp: 29.43°C, Humidity: 32.85%, Pressure: 960.08 hPa, Gas: 38.85 kΩ, AQI: windspeed:0.00 m/s^2
Data sent via WiFi successfully!
```

Figure 9

Serial Monitor shown the Data sending by Wi-Fi connectivity

```
COM6
.....AT+CIPSTART="TCP","api.thingspeak.com","80"
100.79.206.150
.WiFi unavailable, sending data via GSM...
GSM Command: AT+CIPSTART="TCP","api.thingspeak.com","80"
....AT+CIPSTART="TCP","api.thingspeak.com","80"

OK

CONNECT OK
.....GSM Command: AT+CIPSEND
....AT+CIPSEND

> .....GSM Command: AT+CIPSHUT
....GET https://api.thingspeak.com/update?api_key=QBVYPKWKXZ0XTQ&field1=30.11&field2=33.59&field3=960.07&field4=50.09&field5=115&field6=0.00
□
SEND OK
6045AT+CIPSHUT
.....Data sent via GSM successfully!
.....WiFi unavailable, sending data via GSM...
GSM Command: AT+CIPSTART="TCP","api.thingspeak.com","80"
....
SHUT OK
AT+CIPSTART="TCP","api.thingspeak.com","80"

OK
.....GSM Command: AT+CIPSEND
....
CONNECT OK
AT+CIPSEND

> .....GSM Command: AT+CIPSHUT
....GET https://api.thingspeak.com/update?api_key=QBVYPKWKXZ0XTQ&field1=29.92&field2=33.72&field3=960.09&field4=35.09&field5=128&field6=0.00
□
SEND OK
6046AT+CIPSHUT
.....
```

Figure 10

Serial Monitor shown the switching from Wi-Fi to GSM Connectivity


```
COM6
.....WiFi unavailable, sending data via GSM...
GSM Command: AT+CIPSTART="TCP","api.thingspeak.com","80"
....
SHUT OK
AT+CIPSTART="TCP","api.thingspeak.com","80"

OK
.....GSM Command: AT+CIPSEND
....
CONNECT OK
AT+CIPSEND

> .....GSM Command: AT+CIPSHUT
....GET https://api.thingspeak.com/update?api_key=QBVPKWTKEKXZ0XTQ&field1=29.29&field2=36.60&field3=960.68&field4=19.48&field5=143&field6=0.00
□
SEND OK
6117AT+CIPSHUT
.....Data sent via GSM successfully!
.....WiFi unavailable, sending data via GSM...
GSM Command: AT+CIPSTART="TCP","api.thingspeak.com","80"
....
SHUT OK
AT+CIPSTART="TCP","api.thingspeak.com","80"

OK
.....GSM Command: AT+CIPSEND
....
CONNECT OK
AT+CIPSEND

> .....GSM Command: AT+CIPSHUT
....GET https://api.thingspeak.com/update?api_key=QBVPKWTKEKXZ0XTQ&field1=29.29&field2=36.61&field3=960.68&field4=19.44&field5=143&field6=0.00
□
SEND OK
6118AT+CIPSHUT
.....Data sent via GSM successfully!
```

Figure 11

Serial Monitor shown the Data sending by GSM Connectivity

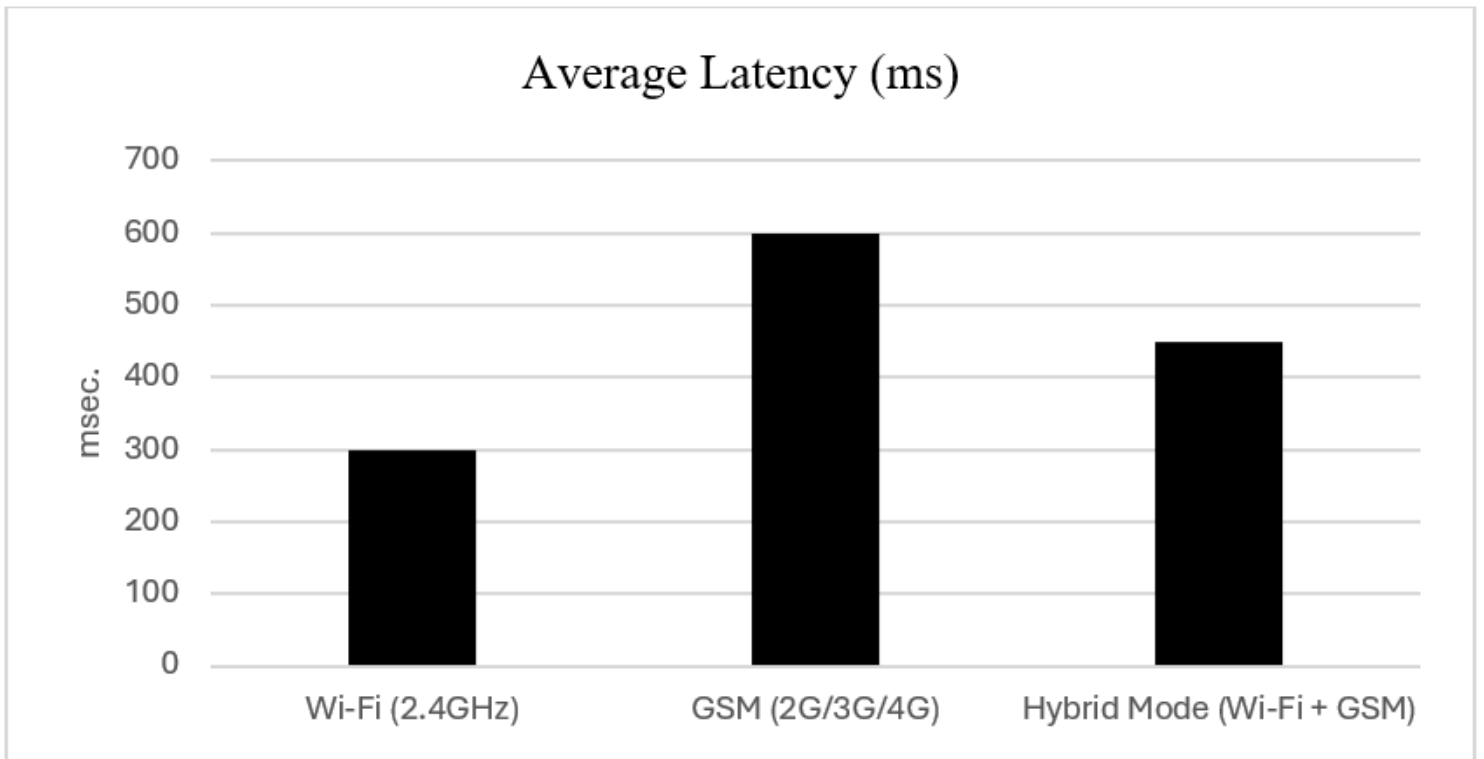


Figure 12

Latency for all modes

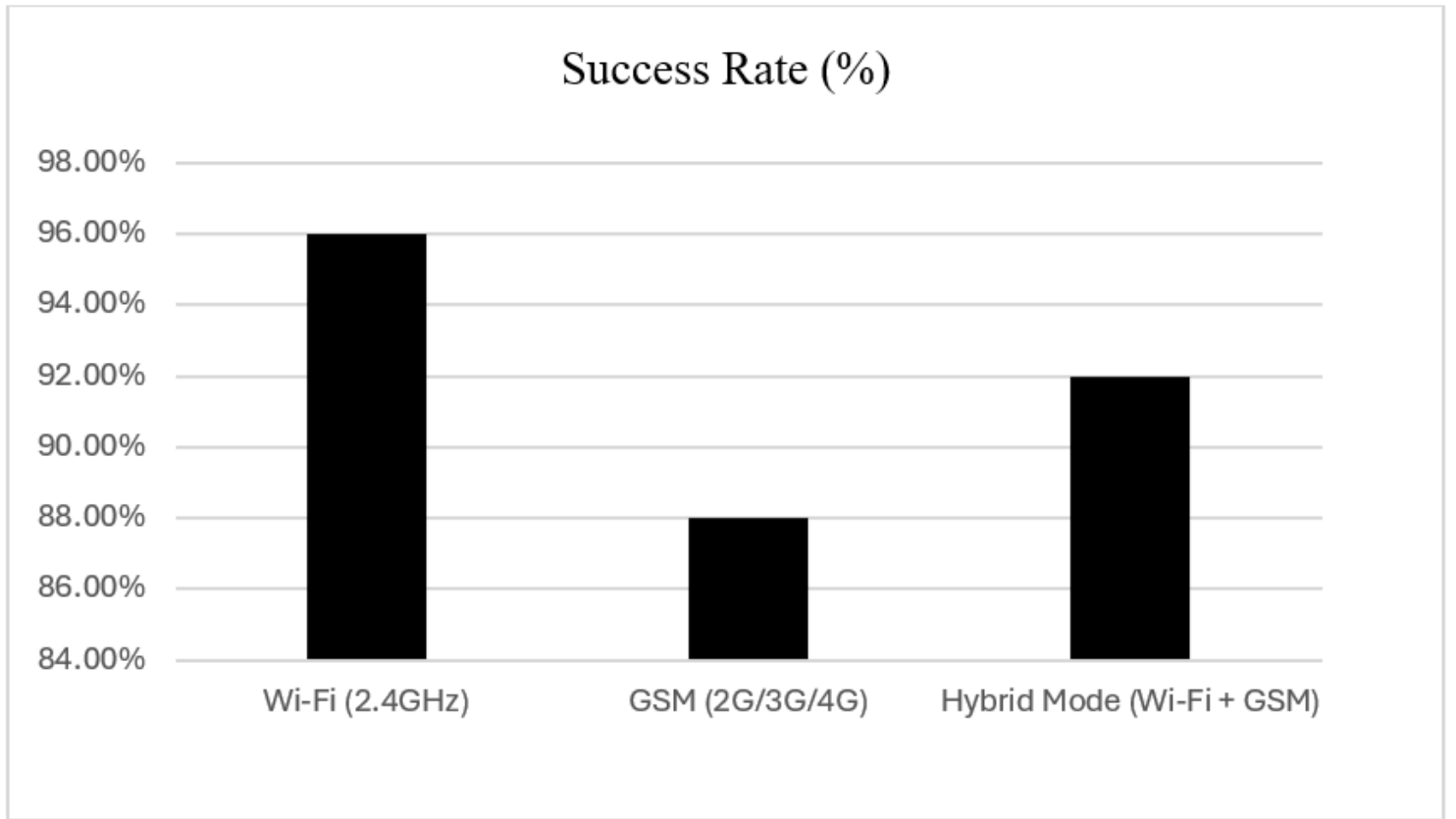


Figure 13

Success rate for all modes

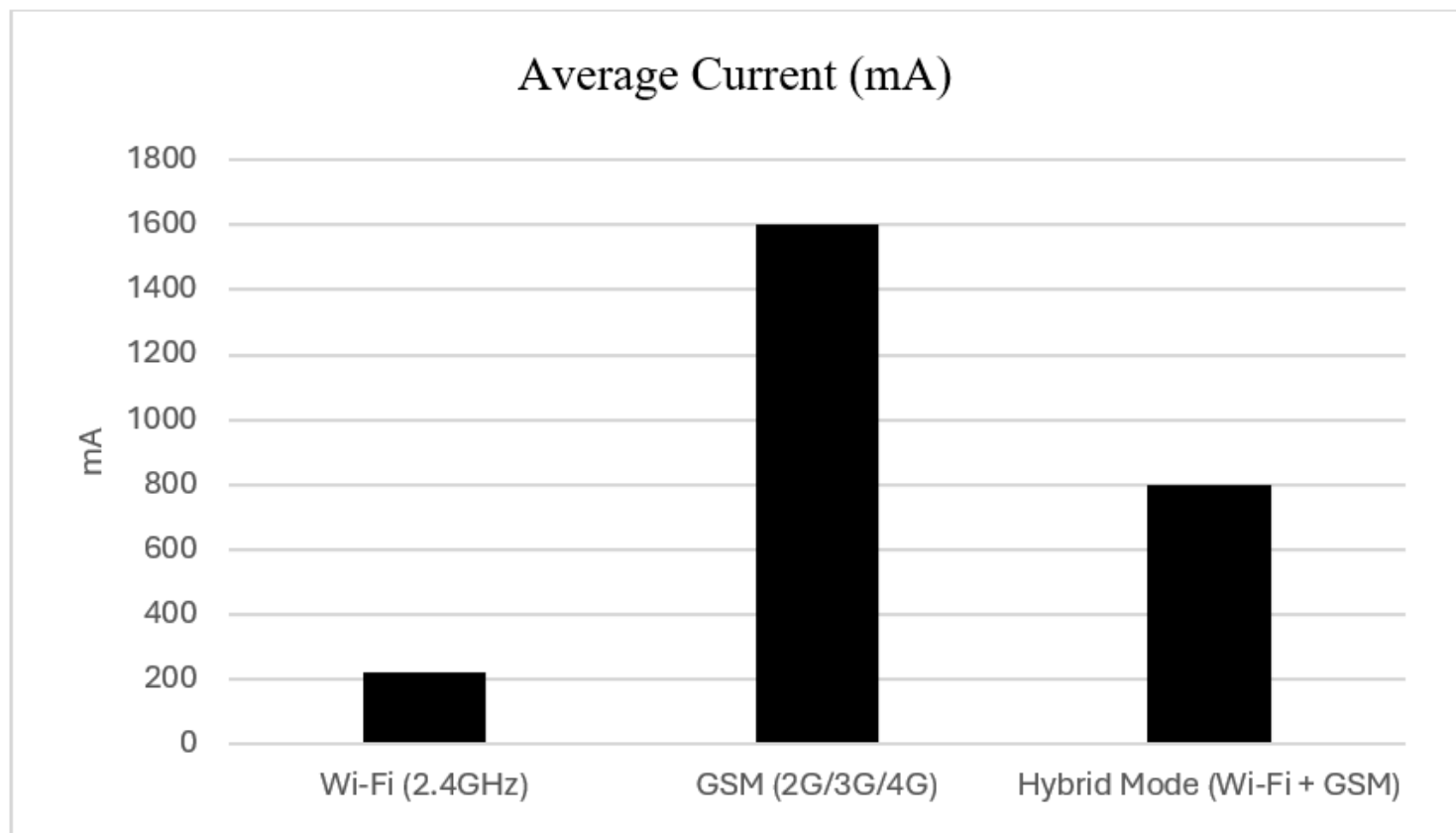


Figure 14

Current Consumption for all modes

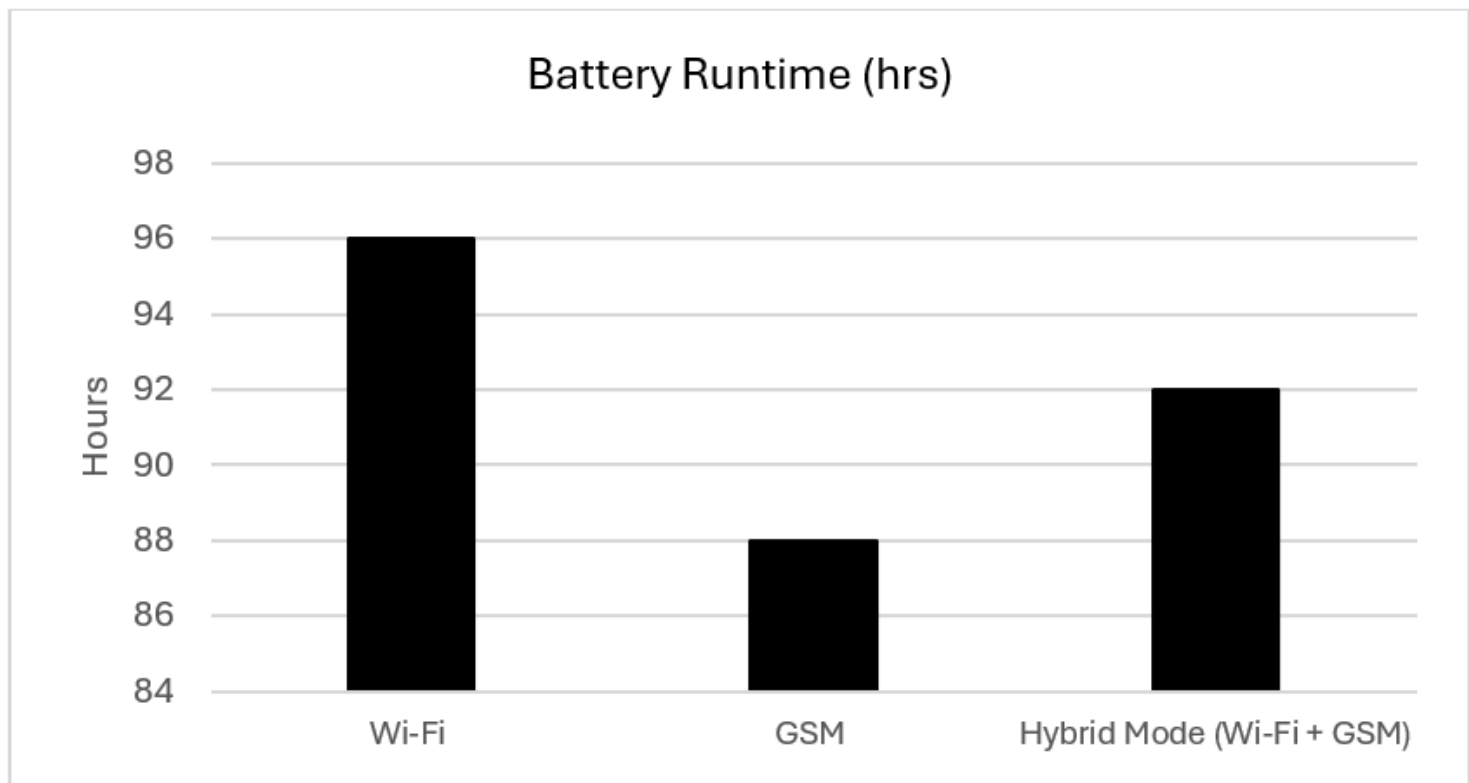


Figure 15

Battery runtime for all modes (*Considering 10000mah battery)

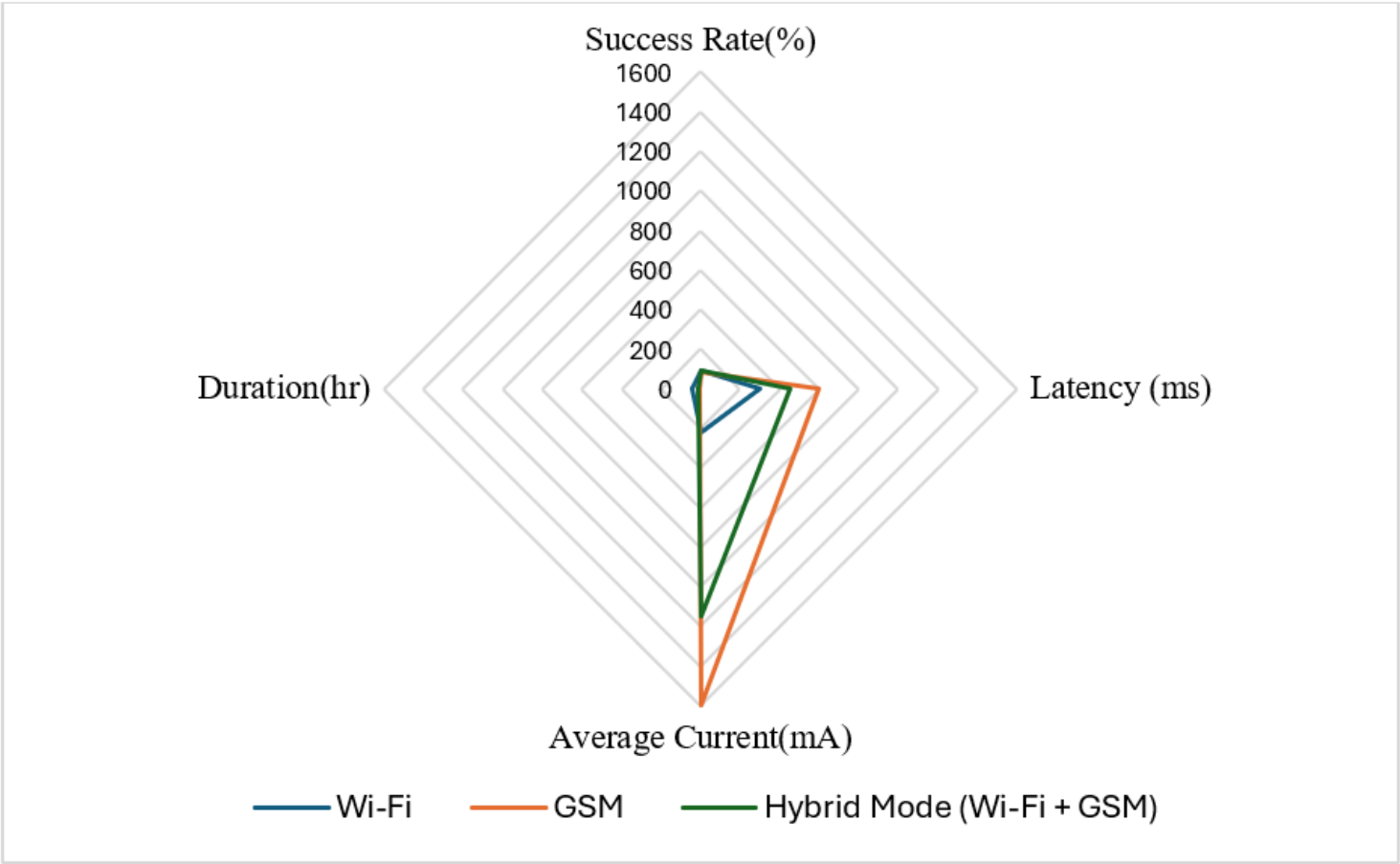


Figure 16

Performance Comparison of between three modes of the proposed Dual-connectivity Data Logger